

ORIGINAL RESEARCH ARTICLE

Association of Pulmonary Vascular Remodeling With Cardiac Structure and Function, Pulmonary Pressure, and Heart Failure in Late Life: The Atherosclerosis Risk in Communities (ARIC) Study

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BACKGROUND: Aging is associated with increases in pulmonary pressure related to concomitant age-associated left ventricular remodeling, diastolic dysfunction, and declines in pulmonary function. Little is known regarding morphological changes in the pulmonary arterial vasculature underlying these associations. Our aim was to determine the associations between pulmonary vascular arterial remodeling, reflected in distal pruning and proximal dilation, and cardiac structure and function, pulmonary pressure, and functional outcomes.

METHODS: Among 2275 participants in the community-based Atherosclerosis Risk in Communities study (ARIC) who underwent echocardiography and noncontrast cardiac computed tomography at study visit 7 (2018–2019), we quantified the fraction of total pulmonary vascular area comprised of arterial vessels with cross-sectional area $<5 \text{ mm}^2$ (aBV5/aTBV) and arterial vessels with cross-sectional area $>10 \text{ mm}^2$ (aBVg10/aTBV) using a validated image processing approach. We assessed the associations of aBV5/aTBV and aBVg10/aTBV with echocardiographic measures of cardiac structure and function and pulmonary artery systolic pressure using multivariable linear regression models adjusted for demographics and cardiovascular risk factors. We also evaluated associations of pulmonary vascular arterial remodeling metrics with circulating NT-proBNP (N-terminal pro-B-type natriuretic peptide), self-reported dyspnea, and incident heart failure.

RESULTS: Mean age was 80 ± 4 years, 61% were women, 22% reported Black race, and mean left ventricular ejection fraction was $64 \pm 7\%$. Mean aBV5/aTBV was 0.30 ± 0.08 , and aBVg10/aTBV was 0.45 ± 0.09 . Lower aBV5/aTBV, reflecting greater distal pruning, and higher aBVg10/aTBV, reflecting greater proximal dilation, were both associated with greater left ventricular remodeling, worse diastolic function, and worse systolic function. Lower aBV5/aTBV and higher aBVg10/aTBV demonstrated nonlinear associations with greater pulmonary artery systolic pressure. Both lower aBV5/aTBV and higher aBVg10/aTBV were associated with higher circulating NT-proBNP and greater odds of moderate to severe dyspnea. Higher aBVg10/aTBV, in particular, was associated with greater risk of incident heart failure over a 4-year follow-up with a hazard ratio of 1.25 (95% CI, 1.03–1.51) per one SD in aBVg10/aTBV.

CONCLUSIONS: Among older adults, pulmonary vascular arterial remodeling is associated with greater left ventricular remodeling, worse diastolic and systolic dysfunction, greater pulmonary artery systolic pressure, greater odds of significant dyspnea, and greater risk of heart failure development. Our findings clarify the morphologic changes in the pulmonary vasculature that link cardiac dysfunction to higher pulmonary pressure in late life and may appear before symptomatology.

GRAPHIC ABSTRACT: A [graphic abstract](#) is available for this article.

Key Words: cohort studies ■ echocardiography ■ hypertension, pulmonary ■ vascular remodeling

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Clinical Perspective

What Is New?

- Left ventricular remodeling, diastolic dysfunction, and systolic dysfunction are associated with greater pulmonary arterial remodeling in late life, characterized by pulmonary arterial distal pruning with proximal dilatation.
- Pulmonary arterial remodeling is nonlinearly associated with pulmonary artery pressure, suggesting a remodeling threshold beyond which pulmonary pressure rises. Pulmonary arterial remodeling is also associated with higher prevalence of participant-reported dyspnea, and higher risk of incident heart failure.

What Are the Clinical Implications?

- Further investigations are warranted to explore pulmonary arterial remodeling, assessed on non-contrast computed tomography imaging, as a risk marker for development of pulmonary hypertension and as a tool to subphenotype pulmonary hypertension attributable to left-sided heart disease.

Advancing age is characterized by changes both in cardiac structure and function and in the pulmonary vasculature. Aging is associated with left ventricular (LV) remodeling,¹ diastolic dysfunction, and concomitant increases in LV diastolic pressure and pulmonary pressure.² Both LV systolic and diastolic dysfunction are associated with greater increases in pulmonary artery systolic pressure (PASP) in late life,³ which, in turn, is associated with heightened risk of heart failure (HF) and mortality: associations that persist in patients >85 years of age.^{4–6} Although meaningful inter-relationships exist between advanced age, pulmonary pressure, and LV remodeling, therapeutic trials targeting pulmonary hypertension (PH) in patients with left heart disease (PH-LHD) have yielded mixed results, and to date, there are no US Food and Drug Administration–approved therapies for PH-LHD. One important barrier is the lack of understanding regarding the specific morphological changes in the pulmonary vasculature underlying the relationship between aging, pulmonary pressure, and cardiac remodeling.

Loss of small vessels, termed pulmonary vascular (PV) pruning, occurs with age,⁷ and histological studies have identified a similar pattern of loss of the distal pulmonary vasculature in cardiac disease and HF with increased pulmonary artery medial wall thickness and decreased luminal diameter.^{8–10} Advanced image processing of non-contrast chest computed tomography (CT) scans¹¹ enables measurement of pulmonary small-vessel volume, the total PV luminal area that is found in small vessels. Most commonly defined as vessels with a cross-sectional

Nonstandard Abbreviations and Acronyms

aBV5	arterial blood volume in vessels <5 mm ²
aBVg10	arterial blood volume in vessels >10 mm ²
AF	atrial fibrillation
ARIC	Atherosclerosis Risk in Communities
aTBV	total arterial blood volume
BMI	body mass index
CAD	coronary artery disease
COPD	chronic obstructive pulmonary disease
CT	computed tomography
EF	ejection fraction
FHS	Framingham Heart Study
HF	heart failure
ICD	<i>International Classification of Diseases</i>
LV	left ventricular
LVEF	left ventricular ejection fraction
mMRC	modified Medical Research Council
PASP	pulmonary artery systolic pressure
PH	pulmonary hypertension
PH-LHD	pulmonary hypertension due to left-heart disease
PV	pulmonary vascular
PVa	pulmonary vascular arterial
RV	right ventricle

intraluminal area <5 mm², this measurement correlates with PV density and remodeling on histology^{12,13} and has been characterized in chronic obstructive pulmonary disease (COPD),^{14–16} bronchiectasis,¹⁷ asthma,¹⁸ chronic thromboembolic PH,¹⁹ acute pulmonary embolus,²⁰ and pulmonary arterial hypertension.²¹ In these contexts, lower pulmonary small-vessel volume (termed pulmonary vascular pruning) correlates with greater right ventricle (RV) size,¹⁶ greater PV resistance, lower cardiac index on right heart catheterization,¹⁹ and worse RV function by cardiac MRI.²² Greater PV pruning is also associated with higher all-cause mortality even in adults without pulmonary disease.²³ The increase in transmural pressure that occurs in conjunction with distal pruning may also lead to dilation of more proximal intraparenchymal vessels,^{24–26} as is seen in PH.^{22,27} Although previous studies have investigated PV remodeling in pulmonary disease, less is known about its associations with cardiac remodeling and dysfunction (the most common causes of PH), especially in the context of advanced age. In this study, we quantified PV remodeling among a community-based cohort of older adults and determined associations with cardiovascular risk factors, subclinical alterations in cardiac structure and function, pulmonary hemodynamics, and clinical outcomes.

METHODS

Study Population

The Atherosclerosis Risk in Communities study (ARIC) is a prospective community-based epidemiological study whose design and methods have been previously described.²⁸ ARIC initially enrolled middle-aged adults between 1987 and 1989 in 4 US communities: Forsyth County, NC; Jackson, MS; suburbs of Minneapolis, MN; and Washington County, MD. The ARIC study protocol was approved by institutional review boards at all participating institutions, and all participants provided written informed consent. This observational study adhered to the STROBE (Strengthening the Reporting of Observational Studies in Epidemiology) reporting guidelines. Anonymized data from the ARIC study are available at the National Heart, Lung, and Blood Institute (NHLBI) Biologic Specimen and Data Repository Information Coordinating Center and can be accessed through the website <https://biolincc.nhlbi.nih.gov/studies/aric/>. Requests for access of ARIC data may also be submitted to the ARIC publications committee according to established study procedures. The code and statistical packages used for analyses in this study are available from the corresponding author upon reasonable request.

Of the 5044 surviving participants, 3065 returned for a seventh visit between 2018 and 2019, at which time they underwent comprehensive echocardiography. Participants were invited to undergo non-contrast cardiac-gated CT scans, for the purpose of CAC scoring, if they did not have clinical coronary artery disease (CAD) based on previous adjudicated myocardial infarction or coronary revascularization by December 31, 2015 (reflecting the lag time of CAD ascertainment in ARIC).²⁹ This analysis included 2275 participants who underwent both echocardiography and non-contrast inspiratory cardiac CT with measurable metrics of PV remodeling (Figure S2).

CT-Based PV Area Reconstruction

Noncontrast cardiac-gated CT scans were performed at full inspiratory breath-hold using multidetector CT scanners by trained technologists and overseen by physicians using identical acquisition protocols.³⁰ Advanced image analysis with generation of 3-dimensional vascular reconstructions (Figure 1) were performed by the Applied Chest Imaging Laboratory (Brigham and Women's Hospital, Boston, MA) using the Chest Imaging Platform (www.chestimagingplatform.org). Intraparenchymal vessels were detected and classified as arteries or veins. Total volume of arterial intraparenchymal vessels (total arterial blood volume [aTBV]), total volume of arterial small vessels (cross-sectional area of $<5 \text{ mm}^2$; arterial blood volume in vessels $<5 \text{ mm}^2$ [aBV5]), and total volume of arterial large vessels (cross-sectional area $>10 \text{ mm}^2$; arterial blood volume in vessels $>10 \text{ mm}^2$ [aBVg10]) were calculated as previously described.²³ The small vessel fraction (aBV5/aTBV) represents the relative volume within the smallest, most peripheral vessels detectable by CT imaging and is an imaging surrogate of PV pruning, with lower values indicating more severe pruning. Measurement of aBVg10/TBV, our metric for proximal intraparenchymal arterial dilation, was chosen as a pragmatic and reproducible metric to capture enlargement of proximal intraparenchymal vessels, rather than as a direct surrogate of strict branching generation based on fractal dimension.²⁷ Larger values indicate more severe dilation.

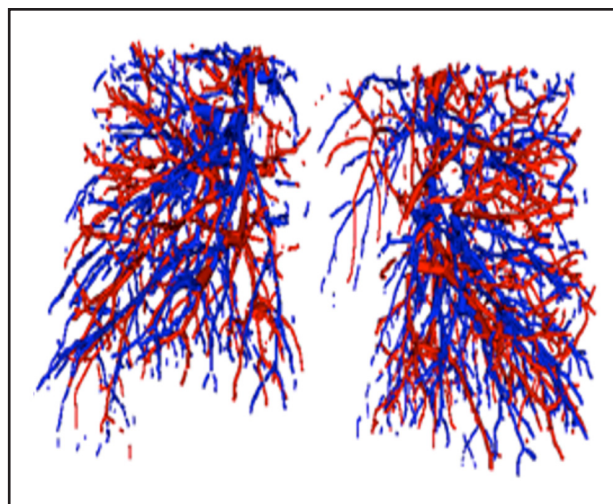


Figure 1. Three-dimensional pulmonary vascular reconstruction from cardiac CT.

Advanced image analysis with generation of 3-dimensional vascular reconstructions were performed by the Applied Chest Imaging Laboratory (Brigham and Women's Hospital, Boston, MA) using the Chest Imaging Platform (www.chestimagingplatform.org). Intraparenchymal vessels were detected and classified as arteries (red) or veins (blue). CT indicates computed tomography.

Whereas previous analyses of small-vessel pruning and proximal intraparenchymal vascular dilation utilized chest CT scans, in this study, we used cardiac CT scans, which exclude portions of the lung fields. To evaluate the validity of using cardiac CT scans for this analysis, we performed a simulation experiment using full inspiratory chest CT scans from the Framingham Heart Study (FHS) and COPDGene (Supplemental Methods). Briefly, we obtained PV volume measurements from a typical cardiac CT region within full chest CT scans and compared them with volume measurements from the entire lung field. We found a high level of correlation between cardiac CT region measurements and full chest CT measurements (Figure S1).

To assess for parenchymal pulmonary disease, low attenuation area percentage was quantified as the percentage of total imaged lung volume that was less than -950 Hounsfield units on noncontrast CT and was used to approximate emphysematous lung changes.³¹ High attenuation area percentage was defined as percentage of total imaged lung volume that was less than -250 Hounsfield units but greater than -600 Hounsfield units on noncontrast CT and was used to approximate fibrotic lung changes.³²

Echocardiography

The design and methods of comprehensive echocardiography in ARIC has been previously reported, including assessments of interobserver variability and temporal drift.³ Briefly, echocardiography was performed at all 4 study field centers by sonographers who were trained and certified in the study-specific imaging protocol. The same echocardiographic machines (Philips iE33, Koninklijke Philips) and probes (Philips XMatrix) were used at all field centers, and all the quantitative measures were performed by a dedicated echocardiography reading center. LV systolic function was assessed based on LV ejection fraction (LVEF), global longitudinal strain and

global circumferential strain. Diastolic measures included early-to-late ventricular filling velocity ratio, tissue Doppler e' , ratio of early mitral inflow velocity to mitral annular velocity, and left atrial (LA) size based on width and volume index. Pulmonary pressure was estimated based on the peak tricuspid regurgitation velocity using the equation: $PASP=4 \times TRV^2+5^{33}$ and was measurable in 1586 participants. RV function was assessed as tricuspid annular plane systolic excursion, RV fractional area change, and tricuspid annular S' .

Assessment of Clinical Covariates

Prevalent hypertension was defined based on self-report of physician diagnosis, medication use, or blood pressure $\geq 140/90$ mm Hg at study visits. Diabetes mellitus was defined from self-report of diagnosis by a physician, medication use, fasting glucose ≥ 126 mg/dL, or nonfasting glucose ≥ 200 mg/dL at study visits. CAD and HF prevalence were based on self-report at visit 1 and surveillance for incident CAD (myocardial infarction and coronary intervention) and HF hospitalization after visit 1 through review of hospital discharge codes with subsequent chart abstraction and committee adjudication, as previously published.^{34–36} Prevalent pulmonary disease was based on validated hospital discharge codes for COPD, asthma, and interstitial lung disease before visit 7 (Supplemental Methods; Table S1).^{37–40} Atrial fibrillation (AF) was ascertained on the basis of hospital discharge *International Classification of Diseases (ICD)* codes and electrocardiograms at study visits 1 through 5.⁴¹ Chronic kidney disease was defined on the basis of an estimated glomerular filtration rate of <60 mL/min/1.73m², calculated using the 2021 Chronic Kidney Disease Epidemiology Collaboration equation. NT-proBNP (N-terminal pro-B-type natriuretic peptide) concentration was measured from serum samples collected at visit 7 using a highly sensitive assay method (Roche Diagnostics),⁴² and values less than the detection limit were assigned a value of 2.5 ng/L.

Dyspnea Assessment

Participant-reported symptoms of dyspnea at visit 7 were evaluated using the modified Medical Research Council (mMRC) scale.⁴³ The mMRC scale score ranged from 0 to 4 on the basis of positive responses to questions regarding the severity of shortness of breath, classified as no (mMRC scale score=0), mild (mMRC scale score=1), moderate (mMRC scale score=2 or 3), or severe (mMRC scale score=4) dyspnea, as previously described.^{3,44} For our analysis, we classified participants as having no (dyspnea score=0, mMRC scale score=0), mild (dyspnea score=1, mMRC scale score=1), or moderate-to-severe (dyspnea score=2, mMRC scale score=2–4) dyspnea.

Incident HF

ARIC participants undergo surveillance for HF through biannual phone questionnaires to identify potential HF hospitalizations and monitoring of local hospitalization discharge *ICD* codes, as described previously,^{34,35} with subsequent chart abstraction and physician adjudication by 2 independent reviewers. The period of follow-up extended from the seventh study visit date through December 31, 2020, for participants at the Jackson study site, and through December 31, 2022, for all other participants. This

was attributable to delays in obtaining hospital records at the Jackson study site.

Statistical Analysis

Echocardiographic variables, aBV5/aTBV, and aBVg10/aTBV were standardized to a mean of 0 and an SD of 1. Summary statistics are presented as means and SD for continuous variables or percentage for binary variables. Low attenuation area percentage and high attenuation area percentage were log-transformed to make the variable distributions near normal. To assess associations with aBV5/aTBV or aBVg10/aTBV, we evaluated baseline clinical characteristics and demographics as well as echocardiographic measures by quartile of aBV5/aTBV or aBVg10/aTBV using linear or logistic regression models as appropriate to evaluate for trend across quartiles. For age, sex, and self-reported race, models adjusted for total imaged lung volume; for total imaged lung volume, models adjusted for age, sex, and race; and for all other baseline clinical characteristics, models adjusted for age, sex, race, and total imaged lung volume.

Associations of echocardiographic measures with pulmonary vascular arterial (PVA) remodeling measures were examined using multivariable linear regression. Two adjustment models were used: model A adjusted for age, sex, race, and total imaged lung volume; model B additionally adjusted for body mass index (BMI), estimated glomerular filtration rate, systolic blood pressure, pack-years of smoking, prevalent CAD, prevalent AF, prevalent hypertension, prevalent diabetes mellitus, low attenuation area percentage, and high attenuation area percentage. To check for nonlinear relationships between measures of PVA remodeling and PASP, we fit restricted cubic splines adjusting for model B covariates. Number of knots minimizing the model Akaike information criterion was selected, with 3 to 5 knots tested. If P for nonlinearity was >0.05 in this model, the association was considered linear. Multivariable logistic regression was used to assess the association of PVA remodeling measures with participant-reported moderate or severe dyspnea, adjusting for model B covariates. Cox proportional hazards regression, adjusting for all model B covariates, was used to assess associations of PVA remodeling measures with time to incident HF among HF-free participants at visit 7. For adjustment covariates, all variables, excluding pack-years, had missingness of $<10\%$, and imputation was performed using multiple imputation by chained equations using 5 imputed data sets and the R package “mice.”⁴⁵ As the primary associations being assessed in this exploratory analysis were 2 related measures of pulmonary arterial remodeling and multiple related measures of cardiac structure and function, we did not view the individual association analyses as independent hypotheses and therefore did not apply a multiple testing penalty. Model assumptions were assessed, including linearity, normality of residuals, heteroskedasticity, and influential points for linear regression models and proportional hazards for Cox proportional hazards models. No violations were observed except for associations of PV measures with PASP, and we used a nonlinear spline method to evaluate these associations.

A 2-sided P value of <0.05 was considered to indicate statistical significance. All analyses were performed using R version 4.4.0⁴⁶ and Stata version 17.⁴⁷

RESULTS

Among the 2275 participants included in this analysis, the mean age was 81 ± 4 years, 61% were women, 22% reported Black race, and cardiovascular risk factors were common (Table 1). Prevalent cardiovascular disease included CAD in 7%, HF in 5%, and AF in 11%; 15% had diagnosed prevalent pulmonary disease. The mean $aBV5/aTBV$ was 0.30 ± 0.08 , and $aBV_{10}/aTBV$ was 0.45 ± 0.09 . Measures of LV systolic function were largely preserved with a mean LVEF $64 \pm 7\%$ and global longitudinal strain of $-18 \pm 3\%$, whereas diastolic indices included a ratio of early mitral inflow velocity to mitral annular velocity of 14 ± 5 , a left atrial volume index of 28 ± 10 mL/m², and a mean PASP within the normal range, at 32 ± 8 mm Hg.

Clinical Correlates of Pulmonary Arterial Remodeling

Male sex, Black race, and greater BMI were all associated with greater distal pruning and greater proximal dilation (Table 1; Table S2). In this late-life cohort, older

age was associated with less distal pruning but was not associated with proximal dilation. Cardiovascular comorbidities, including hypertension, diabetes, AF, and CAD were not associated with distal pruning, whereas diabetes and AF were associated with greater proximal dilation.

LV Structure and Function and Pulmonary Arterial Remodeling

In analyses adjusted for participant characteristics, cardiovascular risk factors, and prevalent cardiovascular disease, measures of LV remodeling, including greater LV wall thickness, end-diastolic volume, and mass index, were each associated with greater small vessel pruning and proximal artery dilatation (Figure 2; Tables S3 and S4). Worse indices of LV diastolic function, including greater left atrium size and early-to-late ventricular filling velocity ratio, were also associated with greater small vessel pruning and proximal artery dilatation. Worse systolic function, reflected by lower LVEF, was associated with greater small vessel pruning and proximal artery dilatation.

Table 1. Baseline Demographics and Clinical Characteristics of the Study Sample at Visit 7 by Quartile of $aBV_{10}/aTBV$.

N	2275	569	569	569	568	P for Trend
	Overall	Q1 (≥ 0.12 , < 0.4)	Q2 (≥ 0.4 , < 0.45)	Q3 (≥ 0.45 , < 0.51)	Q4 (≥ 0.51 , < 0.73)	
Age (y)	80.5 ± 4.3	80.5 ± 4.4	80.8 ± 4.5	80.1 ± 4.1	80.6 ± 4.3	0.73
Women	1395 (61%)	451 (79%)	381 (67%)	324 (57%)	239 (42%)	< 0.001
Non-Black	1772 (78%)	496 (87%)	437 (77%)	431 (76%)	408 (72%)	< 0.001
Prevalent HTN	1989 (87%)	485 (85%)	499 (88%)	496 (87%)	509 (90%)	0.37
Prevalent DM	813 (36%)	174 (31%)	185 (33%)	216 (38%)	238 (42%)	0.01
BMI (kg/m ²)	28.1 ± 5.4	27.0 ± 4.8	27.4 ± 4.9	28.3 ± 5.6	29.8 ± 5.7	< 0.001
Prevalent obesity	701 (31%)	145 (26%)	142 (25%)	178 (31%)	236 (42%)	< 0.001
eGFR (mL/min/1.73m ²)	64 ± 18	62 ± 18	65 ± 17	64 ± 18	62 ± 19	0.85
Prevalent chronic kidney disease	893 (39%)	245 (43%)	207 (36%)	222 (39%)	219 (39%)	0.45
Prevalent AF	251 (11%)	50 (9%)	45 (8%)	72 (13%)	84 (15%)	< 0.001
Prevalent CAD	151 (7%)	29 (5%)	34 (6%)	39 (7%)	49 (9%)	0.16
Prevalent stroke	88 (4%)	20 (4%)	21 (4%)	25 (4%)	22 (4%)	0.70
Prevalent pulmonary disease	278 (15%)	66 (15%)	75 (16%)	63 (13%)	74 (15%)	0.97
Prevalent HF	106 (5%)	15 (3%)	20 (4%)	29 (5%)	42 (7%)	0.002
Pack-years	10.4 ± 18.2	9.0 ± 17.2	9.3 ± 16.4	11.4 ± 19.9	12.1 ± 18.9	0.02
SBP (mm Hg)	135 ± 19	134 ± 19	137 ± 18	134 ± 18	135 ± 20	0.29
DBP (mm Hg)	67 ± 11	65 ± 11	67 ± 10	67 ± 11	68 ± 12	< 0.001
Height (cm)	166 ± 9	162 ± 8	165 ± 9	167 ± 9	170 ± 9	< 0.001
Total imaged lung volume (L)	2.6 ± 0.9	2.5 ± 0.9	2.7 ± 0.9	2.6 ± 0.9	2.5 ± 0.8	< 0.001
LAA950	0.01 [0.00, 0.02]	0.01 [0.00, 0.02]	0.01 [0.00, 0.02]	0.01 [0.00, 0.02]	0.01 [0.00, 0.02]	0.84
HAA	0.08 [0.06, 0.12]	0.09 [0.06, 0.14]	0.07 [0.06, 0.11]	0.08 [0.06, 0.10]	0.08 [0.07, 0.11]	< 0.001

Model is adjusted for age, sex, race, and total imaged lung volume. *P* for trend is linear regression using numeric values assigned to quartiles of PV metrics (quartile 1=1, quartile 2=2, etc). Normally distributed values are reported as mean $\pm$ SD, non-normally distributed values are reported as median [25th–75th percentile]. AF indicates atrial fibrillation; CAD, coronary artery disease; DBP, diastolic blood pressure; DM, diabetes mellitus; HAA, high attenuation area; HF, heart failure; HTN, hypertension; LAA low attenuation area; and SBP, systolic blood pressure.

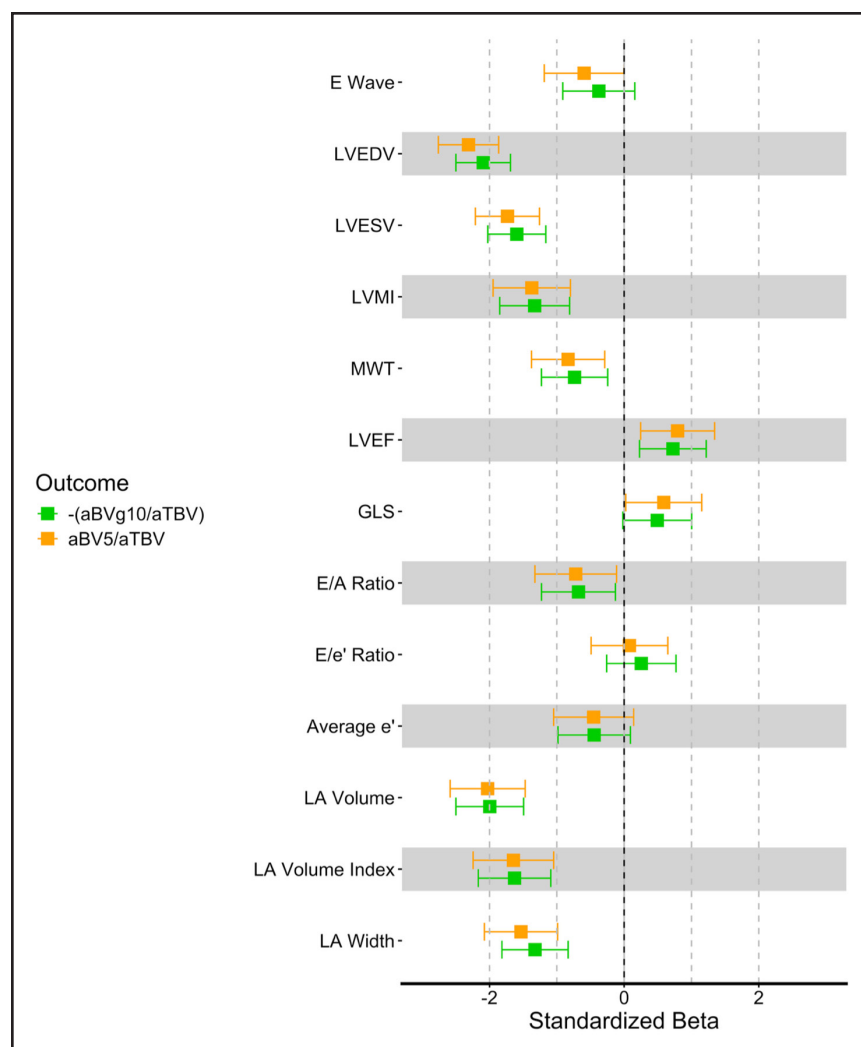


Figure 2. Forest plot of associations of echocardiographic markers of left-heart structure and function with PVA distal pruning and proximal dilation.

Linear models adjusted for age, BMI, eGFR, sex, race, SBP, total imaged lung volume, pack-years smoking, prevalent hypertension, prevalent diabetes mellitus, prevalent atrial fibrillation, prevalent coronary artery disease, low attenuation area percentage, and high attenuation area percentage, as well as heart rate and SBP at time of echo. aBVg10/aTBV indicates large vessel fraction; aBV5/aTBV, small vessel fraction; BMI, body mass index; eGFR, estimated glomerular filtration rate; GLS, global longitudinal strain; LA, left atria; LVEDV, left ventricular end-diastolic volume; LVEF, left ventricular ejection fraction; LVESV, left ventricular end-systolic volume; LVMI, left ventricular mass index; MWT, mean wall thickness; and SBP, systolic blood pressure.

Pulmonary Arterial Remodeling, Pulmonary Pressure, and Right Ventricular Function

Both greater distal pruning (lower aBV5/aTBV) and proximal dilation (higher aBVg10/aTBV) were associated with greater estimated PASP (Figure 3). These associations were nonlinear, such that at higher values of aBV5/aTBV, associations with PASP were negligible, but at lower values of aBV5/aTBV (beginning approximately at values <0.3), lower aBV5/aTBV was associated with higher PASP. Conversely, the association of higher aBVg10/aTBV with higher PASP was primarily observed at higher values of aBVg10/aTBV, beginning approximately at values >0.45. Greater small vessel pruning and proximal artery dilation were both associated with higher tricuspid annular plane systolic excursion and tricuspid annular s', suggestive of greater RV contractility (Table 2).

Biomarkers, Functional Outcomes, and Incident HF

The median NT-proBNP was 165 (25th–75th percentile, 88–321) pg/mL. Greater proximal arterial dilata-

tion was associated with higher NT-proBNP, whereas there was no significant association with distal arterial pruning (Figure 4). Of the 2122 participants with available data on self-reported dyspnea, 12% reported mild dyspnea, and 22% reported moderate or severe dyspnea. Greater proximal pulmonary arterial dilation was associated with severe dyspnea (odds ratio, 1.11 [95% CI 1.01–1.23]; $P=0.04$), although no associations were observed with distal vessel pruning (Table S5). Incident HF occurred in 124 participants over a median follow-up of 4 (25th–75th percentile, 3–4) years. Greater proximal pulmonary arterial dilatation (aBVg10/aTBV) was associated with a greater risk of incident HF (hazard ratio per 1-SD increase, 1.25 [95% CI, 1.03–1.51]; $P=0.02$), with only borderline significant associations with distal vessel pruning (hazard ratio, 1.24 [95% CI, 1.00–1.53]; $P=0.05$; Figure 4).

DISCUSSION

This analysis is one of the first, to our knowledge, to quantify pulmonary arterial remodeling among older

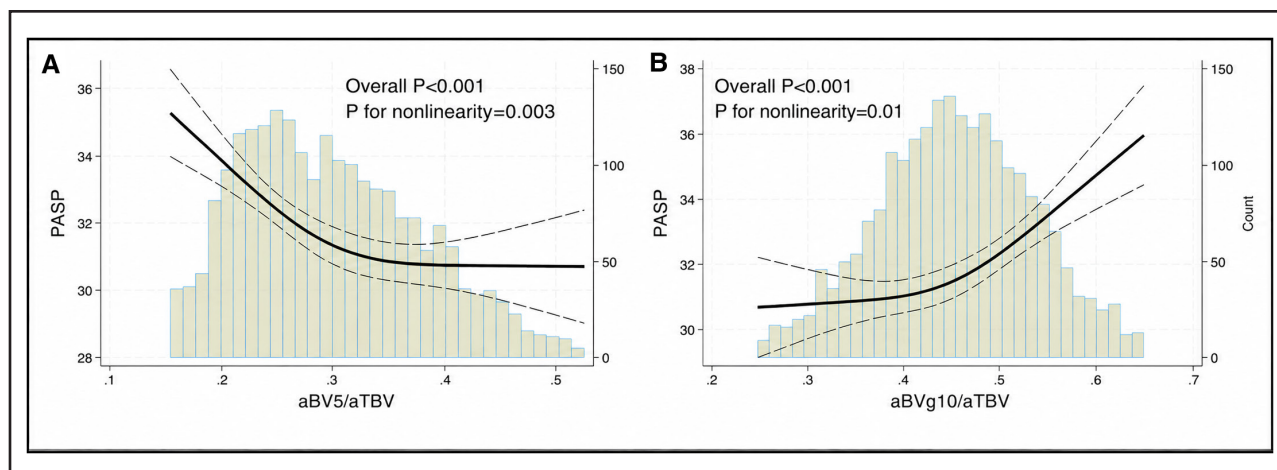


Figure 3. Restricted cubic spline regression showing nonlinear association between PASP and aBV5/aTBV (A) and between PASP and aBVg10/aTBV (B).

Models adjusted for age, BMI, eGFR, sex, race, SBP, total imaged lung volume, pack-years smoking, prevalent hypertension, prevalent diabetes mellitus, prevalent atrial fibrillation, prevalent coronary artery disease, low attenuation area percentage, and high attenuation area percentage. aBVg10/aTBV indicates large vessel fraction; aBV5/aTBV, small vessel fraction; BMI, body mass index; eGFR, estimated glomerular filtration rate; PASP, pulmonary artery systolic pressure; and SBP, systolic blood pressure.

adults in a community-based cohort and to define its associations with cardiac structure and function, pulmonary pressure, participant-reported symptoms, and risk of clinical HF. We report 3 primary novel findings. First, greater LV remodeling, worse diastolic dysfunction, and worse systolic dysfunction are associated with both pulmonary arterial distal pruning and proximal dilation, consistent with left heart dysfunction as the most common etiology of PH. Second, both distal pulmonary arterial pruning and proximal dilation are nonlinearly associated with greater PASP, suggesting a remodeling threshold beyond which pulmonary pressure rises. Third, proximal pulmonary artery dilation, in particular, is associated with higher circulating NT-proBNP, greater frequency of severe dyspnea, and higher risk of incident HF, suggesting clinical ramifications of PV remodeling. Together, these findings suggest that cardiac dysfunction and remodeling in advanced age is associated with PV remodeling, which may occur before clinical PH and HF.

The degree of distal pulmonary arterial pruning was greater in our late-life cohort compared with studies

using similar methodology from largely middle-aged participants in FHS (age 81 ± 4 versus 59 ± 11 years, aBV5/aTBV 0.30 versus 0.39 in ARIC and FHS, respectively).⁴⁸ Few previous analyses have reported aBVg10/aTBV as a measure of proximal vessel dilatation. Although comparisons of results between different studies are challenging, our findings suggest an advanced degree of distal pulmonary arterial pruning in older adults and are supported by a previous study in murine models that showed rarefaction of the pulmonary vasculature, another metric of quantifying distal vascular pruning, in elder mice.⁷ That being said, within our cohort, greater age was associated with less distal vessel pruning. This may represent survivorship bias within our late-life cohort, in which those participants with less distal pruning survived longer. Future studies with serial assessment of PV measurements by CT will be necessary to further elucidate the relationship between aging and PV remodeling.

Among this late-life cohort, male sex and higher BMI were associated with greater distal pulmonary arterial pruning and proximal vessel dilation, whereas Black race

Table 2. Echocardiographic Measures of Right-Ventricular Systolic Function by Quartile of PVa Measurements

PV measure	Echo outcome	Q1	Q2	Q3	Q4	P for trend	Continuous variable P value
aBVg10/aTBV (Higher quartile reflects more PVa remodeling)	TAPSE (cm)	2.04±0.46	2.08±0.47	2.08±0.48	2.12±0.51	0.01	0.01
	TVSA (cm/s)	11.58±2.76	11.89±2.85	12.13±2.85	12.64±3.03	<0.001	<0.001
aBV5/aTBV (Lower quartile reflects more PVa remodeling)	TAPSE (cm)	2.11±0.50	2.10±0.47	2.08±0.49	2.03±0.45	0.001	0.01
	TVSA (cm/s)	12.66±2.95	12.34±2.87	11.69±2.76	11.52±2.85	<0.001	<0.001

Models adjusted for age, body mass index (BMI), eGFR, sex, race, SBP, total imaged lung volume, pack-years smoking, prevalent hypertension, prevalent diabetes mellitus, prevalent atrial fibrillation, prevalent coronary artery disease, low attenuation area percentage, and high attenuation area percentage, as well as heart rate and SBP at time of echo. P for trend is linear regression using numeric values assigned to quartiles of PV metrics (quartile 1=1, quartile 2=2, etc). Continuous variable P value uses PV measures as continuous variables. Refer to Table 1 and Table S2 for quartile definitions for aBVg10/aTBV and aBV5/aTBV, respectively. aBVg10/aTBV indicates large vessel fraction; aBV5/aTBV, small vessel fraction; PVa, pulmonary vascular arterial; TAPSE, tricuspid annular plane systolic excursion; and TVSA, tricuspid valve annular S'.

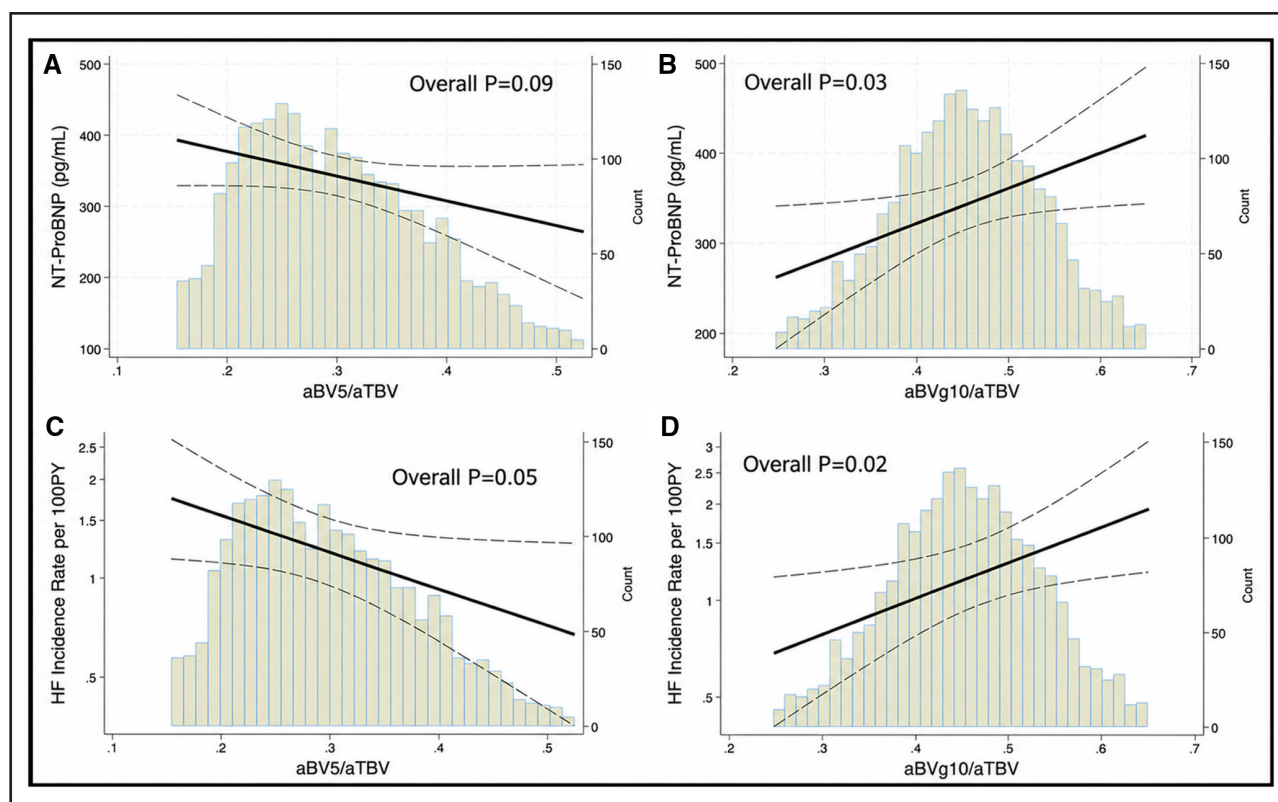


Figure 4. Associations between PVA measurements and NT-proBNP (A and B) and incident HF (C and D).

Models adjusted for age, BMI, eGFR, sex, race, SBP, total imaged lung volume, PY smoking, prevalent hypertension, prevalent diabetes mellitus, prevalent atrial fibrillation, prevalent coronary artery disease, low attenuation area percentage, and high attenuation area percentage. Nontransformed values of NT-proBNP were used for statistical models. aBVg10/aTBV indicates large vessel fraction; aBV5/aTBV, small vessel fraction; BMI, body mass index; eGFR, estimated glomerular filtration rate; HF, heart failure; NT-proBNP, N-terminal pro-B-type natriuretic peptide; PASP, pulmonary artery systolic pressure; PY, pack-years; and SBP, systolic blood pressure.

was associated with greater distal pruning alone. These findings are in agreement with, and extend upon, previous data from individuals with COPD enrolled in the COPDGene study⁴⁹ and younger community-based participants in FHS. The reason for different patterns of distal pruning and proximal dilation based on race and sex is not entirely clear. Conversely, the association of obesity with PH is well established.⁵⁰ Multiple proposed mechanisms may explain the relationship between higher BMI and greater PV remodeling, including obesity-associated systemic inflammation, insulin resistance, and oxidative stress,⁵¹ which can result in endothelial dysfunction and vascular remodeling.⁵²

Notably, the extent of distal pruning was not associated with other cardiovascular risk factors in our analysis, whereas greater proximal dilation was associated with diabetes, AF, and cumulative smoking pack-years. Histological studies have found thicker pulmonary arteriolar and capillary walls in patients with diabetes compared with nondiabetic controls,^{53,54} consistent with our findings and suggesting that proximal vascular dilation on CT may be a more specific marker of PV remodeling compared with distal vessel pruning. Previous studies have shown that tobacco exposure is counterintuitively asso-

ciated with less distal pruning,⁵⁵ despite being known to cause smooth muscle proliferation and intimal thickening pathologically,^{56,57} suggesting that distal pruning measured by CT may not be a useful tool for measuring these pathological changes. Our findings reinforce that distal vessel pruning measured radiographically is not associated with tobacco exposure. However, our observation of an association with proximal dilation also suggests that proximal dilation may be a more sensitive marker of pathological remodeling. The absence of associations of PVA with most cardiovascular risk factors, including hypertension, may be partially explained by the age of our cohort and high prevalence of these comorbidities in late life. Further studies are needed to clarify whether cardiovascular risk factors are associated with PVA remodeling in younger individuals. Additionally, our study was limited in its ability to evaluate associations between PVA remodeling and CAD, as participants with pre-existing CAD before December 31, 2015, were excluded. Further studies including a greater proportion of participants with prevalent CAD are needed to further elucidate the association between PVA remodeling and CAD.

Greater LV remodeling and worse diastolic function were associated with greater pulmonary arterial

distal pruning and proximal dilation. In our cohort with predominantly preserved systolic function (4% with an LVEF <50%), systolic measures were not associated with pulmonary arterial pruning. These findings are consistent with previous histological studies demonstrating increased medial wall thickness in both pulmonary arteries and veins in HF, and potentially of greater magnitude in HF with preserved EF compared with HF with reduced EF.^{9,10,58} Although these studies focused on patients with advanced HF, our findings now extend these observations to a cohort that is predominantly HF-free (92% without clinical HF) but at increased HF risk because of age and clinical risk factors. Our findings demonstrate that PV remodeling occurs in the setting of subclinical LV remodeling and diastolic dysfunction, before clinical signs and symptoms.

In turn, greater pulmonary arterial pruning and proximal dilatation were associated with higher PASP, consistent with previous studies.^{21,24,59} Importantly, we observed a nonlinear relationship between the extent of PVa remodeling and PASP, suggesting that pulmonary pressure remains unaffected with early PVa remodeling up to a threshold beyond which pulmonary pressure rises linearly with further distal pruning and proximal dilation. This inflexion point, or threshold, occurred in our analysis at a small vessel fraction of 0.30 and large vessel fraction of 0.45 (Figure 3). This finding is biologically plausible, as previous studies have found that histological remodeling in the pulmonary vasculature precedes elevations in pulmonary pressure in certain etiologies of PH.^{60,61} The clinical implication of the nonlinear relationship between PVa remodeling and PASP is that CT-derived markers of vascular remodeling may be useful for early risk-stratification of patients before elevations in pulmonary pressure.

Together, these findings suggest that noninvasively assessed PV pruning is a potential early marker of risk for elevated pulmonary pressure and PH and may be particularly useful as an early marker of PH-LHD. This is clinically important, as early diagnosis of PH is associated with improved survival⁶² but remains challenging to diagnose clinically, as symptoms may not present until late in the disease course.⁶³ A validated noninvasive marker of elevated risk for PH-LHD from a widely available modality such as CT could therefore be helpful for modifying the disease course of this highly prevalent and morbid condition. Furthermore, noninvasive metrics of PV morphology may be helpful in subphenotyping patients with PH-LHD. There is a subset of patients with PH-LHD with elevated pulmonary pressure out of proportion to left-sided cardiac filling pressure, thought to be attributable to PV remodeling and termed combined precapillary and postcapillary PH. These patients have a worse prognosis than those with isolated postcapillary PH.^{64,65} To date, therapies targeting the pulmonary vasculature have not demonstrated consistent benefit in postcapillary PH;⁶⁶ however, those with greater degrees of mor-

phological PV remodeling on noninvasive imaging may represent a subpopulation more likely to benefit from PH interventions specifically targeting vascular remodeling.⁶⁷ Further study is needed to identify whether CT-based measures of PV remodeling are associated with greater clinical benefit from PH interventions.

Previous studies have demonstrated a relationship between greater PV pruning and worse right ventricular size and function in the setting of acute submassive pulmonary embolism and COPD.^{16,68,69} We now extend these findings to older adults in the community free of acute illness. In contrast to previous studies, greater distal PV pruning and proximal dilation were associated with higher tricuspid annular plane systolic excursion and tricuspid s' in our study, suggesting associations of greater PVa remodeling with increased RV systolic function. The most likely explanation for this finding is RV-PA coupling, in which RV contractility increases in step with increased RV afterload in the pulmonary circulation to maintain cardiac output.⁷⁰ RV-PA uncoupling, on the other hand, occurs when the PA pressures are so great that RV contractility cannot adequately adapt to match the excess afterload, which likely explains the findings of the previous analyses in the setting of acute pulmonary embolus and COPD.⁶⁸

Finally, our analysis revealed multiple clinical ramifications of PV remodeling. PVa remodeling measures were associated with higher NT-proBNP levels, greater participant reported moderate or severe dyspnea, and greater risk of incident HF over a 4-year follow-up. Numerous studies have established the prognostic value of distal PV pruning in a number of pulmonary disease states, including COPD, bronchiectasis, acute pulmonary embolus, chronic thromboembolic PH, and pulmonary arterial hypertension. Interestingly, previous data from FHS found that greater distal pulmonary arterial pruning was associated with increased risk of all-cause mortality in a general population.²³ Our findings further support that PV remodeling measured by CT is a useful prognostic marker outside of pulmonary disease, and is the first to our knowledge to describe an association between noninvasive measures of PV remodeling and HF-risk. Notably, the generally stronger associations of PVa measures of proximal vessel dilatation compared with distal vessel pruning with these outcomes suggest potentially greater sensitivity of proximal pulmonary arterial dilatation as a marker of clinically important remodeling of the pulmonary arterial bed and warrants further investigation.

This analysis has several limitations. PASP was measured noninvasively using transthoracic echocardiogram as opposed to direct measurement with right heart catheterization, which introduces error.⁷¹ Of the 2275 ARIC participants at visit 7 who underwent echocardiography and had CT-based PV measures assessed, 766 (34%) did not have a measurable peak tricuspid regurgitation velocity or PASP. Missing PASP on transthoracic echocardiogram is nonrandom and associated with a higher

BMI and lower invasively measured mean pulmonary artery pressure, potentially introducing bias into analyses of pulmonary arterial metrics with PASP.^{72,73} We used a cohort of study participants with advanced age, with a mean age of 81 years. Although our findings are specifically informative about the relationship among PV remodeling, pulmonary pressure, and cardiac function in advanced age, these findings may not be generalizable to younger populations. aBVg10/TBV assesses enlargement of proximal intraparenchymal arteries, as opposed to the anatomically central vasculature, and is not a surrogate of strict branching generation. Future studies should evaluate alternative anatomical definitions of centrality in data sets with full lung coverage; however, the primary hypothesis of this study was that early vascular remodeling is first detectable within the intrapulmonary vasculature, preceding measurable involvement of the more proximal central vessels. The cross-sectional design does not allow for assessment of the temporal sequence between PV remodeling, pulmonary hemodynamics, and cardiac structure and function. Therefore, we are unable to make assertions about whether cardiac remodeling antecedes PV remodeling. Furthermore, we are unable to determine causal relationships. Next, although decreased small vessel fraction has been shown to correlate with PV pruning on histology,¹³ these noninvasive metrics may result in misclassification of histological PV remodeling. Finally, all CT scans analyzed were acquired using a standardized cardiac CT field-of-view, defined by cardiac anatomy rather than pulmonary landmarks. As such, the imaged thoracic volume was determined by the cardiac acquisition protocol, not by disease status or operator discretion, minimizing the risk of an intrinsic technical bias related to acquisition parameters across participants. However, anatomical interindividual variability in cardiac position within the mediastinum can result in modest differences in the specific lung regions included across subjects and may lead to preferential undersampling of more distal or peripheral lung regions in some individuals, specifically in the upper lobes. However, in our simulation experiment using full chest CT scans from 2 large, population-based cohorts (COPDGene and FHS), we compared pulmonary volume measurements from approximated cardiac CT fields with those from full lung fields and found a high degree of correlation for both aBV5 and aTBV. Although these findings suggest that measurements from cardiac CTs are highly representative of the entire lung field, future studies using complete lung imaging would be valuable to replicate these findings.

Conclusions

LV remodeling, diastolic dysfunction, and systolic dysfunction are associated with greater PVA remodeling

in late life. In turn, PVA remodeling is associated with greater pulmonary artery pressure and worse outcomes, including participant-reported dyspnea and risk of incident HF. Further investigations are warranted to explore pulmonary arterial remodeling, assessed on noncontrast CT imaging, as a risk marker for development of PH and as a tool to subphenotype PH-LHD.

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Disclosures

None.

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